

Typescript Programs:

1. WAP to Reverse the String without any functions

```
function reverseString (str : string) : string {  
    let revString: string = '';  
    for (let i = str.length-1; i >= 0; i--){  
        revString +=str[i];  
    }  
    console.log(revString);  
    return revString;  
}  
reverseString('payal');
```

2. WAP to reverse string with function

```
let str: string = 'Try programiz.pro';  
let reverseStr: string = str.split('').reverse().join('');  
console.log(reverseStr);
```

3. WAP to find duplicate in array

```
const arr = [1,2,3,2,4,2,1,4,5];  
const count = {};  
const dupltArr = [];  
  
for (const item of arr) {  
    count[item] = (count[item] || 0) + 1;  
}  
for (const key in count) {  
    if (count[key] > 1) {  
        dupltArr.push(key);  
    }  
}  
  
console.log(dupltArr);
```

Output: [1, 2, 4]

4. WAP to find longest string from the sentence

```
function findLongestWord(str: string) : string {  
  
    if(str.trim().length === 0){  
        return str;  
    }  
  
    const words: string[] = str.split(" ");  
    //console.log(words);  
    let longest: string = " ";  
  
    for(const word of words){  
        if(word.length > longest.length){  
            longest = word;  
        }  
    }  
    return longest;  
}  
  
console.log(findLongestWord("I love programming in TypeScript"));
```

5. WAP to find palindrome number

```
function isPalindrome(num: number): string{  
    let palIndmNum = "";  
    let newnumber = num.toString();  
    for(let i = newnumber.length-1; i >= 0; i--){  
        palIndmNum += newnumber[i]  
    }  
    if(palIndmNum == newnumber){  
        console.log(`Number is Palindrome Number` +palIndmNum)}  
    else{  
        console.log(`Number is not Palindrome Number` +palIndmNum)}  
    return palIndmNum;  
}
```

```
}
```

```
isPalindrome(122);
```

6. WAP to generate hash tag for given string and string should be less than 270 and not white spaces

```
function hashTagGenration(str){
```

```
  if (str.trim().length ===0 || str.length > 230)
```

```
  {
```

```
    return false;
```

```
  }
```

```
  str = str.split(' ');
```

```
  str = str.map((curElem) =>
```

```
    //curElem.replace(curElem[0], curElem[0].toUpperCase())
```

```
    curElem.charAt(0).toUpperCase() + curElem.slice(1)
```

```
  );
```

```
  str = `#${str.join("")}`
```

```
  console.log(str);
```

```
  return str;
```

```
};
```

```
hashTagGenration(`My new house is beatiful`);
```

7. WAP to find occurrence of each character in String

```
function getCharCount(str){
  charCount = {};
  for(let char of str){
    char = char.toLowerCase();
    if(charCount[char]){
      charCount[char]++;
    }else{
      charCount[char]=1;
    }
  }
  return charCount;
}
console.log(getCharCount('Selenium'));
```

8. WAP to convert the strings of array in uppercase

Method : 1

```
let colors = ["orange", "blue", "green"];

for(let i = 0 ; i <= colors.length-1; i++){
  colors[i] =colors[i].toUpperCase();
}

console.log(colors);
```

Method: 2

```
let fruits = ["apple", "banana", "mango", "orange"];

let upperFruits = fruits.map(fruits => fruits.toUpperCase());

console.log(upperFruits);
```

9. Find even number from array

Method : 1

```
let numbers = [1,2,3,4,5,6,7,8,9,10]
```

```
let evenNum = numbers.filter(numbers => numbers % 2 === 0)
```

```
console.log(evenNum)
```

Method : 2

```
let evenNumbers = [];
```

```
for(let i= 1; i < numbers.length; i++){  
  if (numbers[i] % 2 === 0){  
    evenNumbers.push(numbers[i]);  
  }  
}  
console.log(evenNumbers)
```

10.Remove duplicates from array

```
let arr = [1,2,2,3,4,5,4,3]
```

```
//let unique = [...new Set (arr)]
```

```
//console.log(unique);
```

```
let uniqueArr = [];  
for(let i = 0; i<arr.length; i++){  
  if(!uniqueArr.includes(arr[i])){  
    uniqueArr.push(arr[i])  
  }  
}  
console.log(uniqueArr);
```

output: [1,2,3,4,5]

11.Reverse the word from given string

```
let str = "I love my India"
```

```
str = str.split(' ').reverse().join(' ');
```

```
console.log(str);
```

Output: India my love I

Method:2

```
let str = "I love Payal";
```

```
let words = str.split(" ");
```

```
let reverse = " ";
```

```
for(let i = words.length-1; i>=0 ; i--){
```

```
    reverse += words[i] + " ";
```

```
}
```

```
console.log(reverse);
```

Output: Payal love I

12.Prgramme input: "I love Payal" and

output: I Evol Layap

```
let str = "I love Payal";
```

```
let words = str.split(" ");
```

```
let result = words.map(word => {
```

```
    let reversed = "";
```

```
    for (let i = word.length - 1; i >= 0; i--) {
```

```
        reversed += word[i];
```

```
    }
```

```
    return reversed.charAt(0).toUpperCase() +  
    reversed.slice(1).toLowerCase();  
});
```

```
console.log(result.join(" "));
```

13.Flatterring the array:

input : let arr = [1,2,[3],[4,5]]; output ://[1,2,3,4,5]

```
let arr = [1, 2, [3], [4, 5]];
```

```
function flattenArray(input: any[]): any[] {  
  let result: any[] = [];
```

```
  for (let i = 0; i < input.length; i++) {  
    if (Array.isArray(input[i])) {  
      let flattened = flattenArray(input[i]);  
      for (let j = 0; j < flattened.length; j++) {  
        result.push(flattened[j]);  
      }  
    } else {  
      result.push(input[i]);  
    }  
  }  
}
```

```
  return result;  
}
```

```
let output = flattenArray(arr);  
console.log(output); // [1, 2, 3, 4, 5]
```

Method 2

If built-in is allowed (simplest):

```
let output = arr.flat(Infinity);  
console.log(output);
```

14.WAP for below input and output

Input : let arr = [2, 0, 0, 3, 0, 0, 1, 0, 0, 5, 0];

Output: [2, 3, 1, 5, 0, 0, 0, 0, 0, 0]

Method 1:

```
let arr2 = [];  
let arr3 = [];
```

```

for(let i = 0; i<=arr.length-1; i++){
  if(arr[i] !== 0){

    arr2.push(arr[i]);

  }else{
    arr3.push(arr[i]);
  }
}

console.log(arr2+" "+arr3);

```

Method 2:

```

let arr = [2, 0, 0, 3, 0, 0, 1, 0, 0, 5, 0];

function moveZeros(input: number[]): number[] {
  let result: number[] = [];
  let zeroCount = 0;

  // Step 1: collect non-zero elements
  for (let i = 0; i < input.length; i++) {
    if (input[i] !== 0) {
      result.push(input[i]);
    } else {
      zeroCount++;
    }
  }

  // Step 2: add zeros at the end
  for (let i = 0; i < zeroCount; i++) {
    result.push(0);
  }

  return result;
}

```



```
let output = moveZeros(arr);  
console.log(output); // [2, 3, 1, 5, 0, 0, 0, 0, 0, 0]
```

15. Program Asked in Coforge

```
let arr = [1,2,1,2,2,4,5,5,2,1,5];  
  
let countMap = {};  
  
for (let num of arr) {  
    countMap[num] = (countMap[num] || 0) + 1;  
}  
  
for (let key in countMap) {  
    console.log(`number ${key}: ${countMap[key]}`);  
}
```

Ouptput:

```
number 1: 3  
number 2: 4  
number 4: 1  
number 5: 3
```

16. Unique and count of number from array

```
let arr = [1,2,1,2,2,4,5,5,2,1,5];  
  
let countMap = {};  
let uniqueArr = [];  
  
for (let num of arr) {  
    // Count  
    countMap[num] = (countMap[num] || 0) + 1;
```

```

// Unique
if (countMap[num] === 1) {
    uniqueArr.push(num);
}
}

console.log("Unique:", uniqueArr);
console.log("Count:", countMap);

```

output:

Unique: [1, 2, 4, 5]
 Count: {1: 3, 2: 4, 4: 1, 5: 3}

17. Check a single number is prime number or not

```

function isPrime(num) {
    if (num <= 1) return false;

    for (let i = 2; i <= Math.sqrt(num); i++) {
        if (num % i === 0) {
            return false;
        }
    }
    return true;
}

```

```

// Example
console.log(isPrime(7)); // true
console.log(isPrime(10)); // false

```

18. Find all prime numbers up to N

```

function getPrimes(n) {
    let primes = [];

```

```

    for (let i = 2; i <= n; i++) {
        if (isPrime(i)) {
            primes.push(i);
        }
    }

    return primes;
}

// Example
console.log(getPrimes(20)); // [2, 3, 5, 7, 11, 13, 17, 19]

```

22. Check Anagram

```

function areAnagram(str1, str2) {
    if(str1.length !== str2.length )
        return console.log(`Words ${str1} and ${str2} are not Anagarm`);
    //return false;
}

let sortedstr1 = str1.split('').sort().join('');
let sortedstr2 = str2.split('').sort().join('');
// console.log(sortedstr1);
// console.log(sortedstr2);
if(sortedstr1 === sortedstr2){
    return (console.log(`Words ${str1} and ${str2} are Anagarm`));
    //return true;
}else{
    return console.log(`Words ${str1} and ${str2} are not Anagarm`);
    //return flase;
}

```

```

}
areAnagram("listen", "silent");
//console.log(areAnagram("listen", "payal")); //false

```

23. First NonRepeated Char from given string

```

function firstNonRepeatedChar(str){
let nonRepeatedChar = "";
  for(let i =0; i<str.length; i++){
    for(let j= 0; j<str.length; j++){
      if(str[i] != str[j]){
        nonRepeatedChar = str[j];
      }
    }
  }
  return nonRepeatedChar;
}
console.log(firstNonRepeatedChar("abcddbec"));

```

output: e

24. Find missing element from array

```

const findMissigNum = (arr) => {
  let total = ((arr.length+1)*(arr.length+2))/2;
  for(let i=0; i<arr.length; i++){
    total = total - arr[i];
  }
}

```

```
    return total;
};
console.log(findMissigNum([1,2,3,4,5,7]));
```

OutPut: 6

25. Second Largest from given array

```
let arr =[1,4,5,7,3,2,9,8];
let largest = "";
let secondlarg="";
for(let i=0; i<arr.length; i++){
    if(largest < arr[i]){
        secondlarg = largest;
        largest = arr[i];
    }
    if(secondlarg < arr[i] && largest > arr[i]){
        secondlarg = arr[i]
    }
}
```

```
console.log(secondlarg);
```

26. Find common number from array

```
function commanNum(arr1, arr2){
    let common = [];
    for(let i=0; i<arr1.length;i++){
        for(let j =0; j<arr2.length; j++){
```

```

        if(arr1[i] === arr2[j]){
            common.push(arr1[i])
        }
    }
}
return common;
}

console.log(commanNum([1,5,10,15,20],[2,10,5,15,11]));
OutPut: 5,10,15

```

27. Merge and sort 2 array

```

const mergSortArray = (arr1, arr2)=>{
    let result = [...arr1,...arr2].sort((a,b)=> a-b);
    return result;
}

console.log(mergSortArray([1,3,2,4], [5,7,6,8]));

```

28. Find vowels in string

```

const countVowels = (str)=>{
    let count = 0;
    let vowels=['a','e','i','o','u'];
    for(let char of str.toLowerCase()){
        if(vowels.includes(char)){
            count++;
        }
    }
}

```

```
    return count;
}
console.log(countVowels('My name is Payal Thakur'))
```

Output : 7

29. Find array is sorted or not

```
const isArraysorted= (arr)=>{
    let sorted = true;
    for(let i = 1; i<arr.length; i++){
        if(arr[i]<arr[i-1]){
            sorted =false;
            break;
        }
    }
    console.log(sorted);
}
isArraysorted([1,2,5,4,3,1]);
```

Output: False

30. Find count of words from string

```
let str = "Payal is QA. Payal is having a good day";
let words = str.toLowerCase().replace(/^[^w\s]/g, "").split(" ");
let wordCount = {};
for (let word of words) {
    wordCount[word] = (wordCount[word] || 0) + 1;
}
```

```
console.log(wordCount);
```

```
output: { payal: 2, is: 2, qa: 1, having: 1, a: 1, good: 1, day: 1 }
```

31. Count duplicate Characters from the string

```
let str = 'Selenium';
```

```
str = str.toLowerCase();
```

```
let countDup = {};
```

```
let duplicate = [];
```

```
for(let char of str){
```

```
    countDup[char] = (countDup[char] || 0)+1;
```

```
}
```

```
for(let key in countDup){
```

```
    if(countDup[key]>1){
```

```
        duplicate.push(key)
```

```
        console.log(`Duplicate Character is = ${duplicate}: and count is =  
        ${countDup[key]}`);
```

```
    }
```

```
}
```

Output: Duplicate Character is = e: and count is = 2

32. ABS company asked below program

```
let str = "String- this is a test this is simple this";
```

```
str = str.toLowerCase();
```

```
let words = str.split(" ");
```

```
let count = {};
```

```
for (let word of words) {
```

```
    count[word] = (count[word] || 0) + 1;
```

```
}
```



```
const sorted = Object.entries(count)
```

```
.sort((a, b) => b[1] - a[1]);
```

```
console.log(sorted);
```

Output: [['this', 3], ['is', 2], ['string-', 1], ['a', 1], ['test', 1], ['simple', 1]]

33. Marvic Solution asked below program:

```
let input = "A1B2C3D4E5";
```

```
let alphabets = "";
```

```
let numbers = "";
```

```
let vowels = "";
```

```
for (let char of input) {
```

```
  if (/[a-zA-Z]/.test(char)) {
```

```
    alphabets += char;
```

```
    if ("aeiouAEIOU".includes(char)) {
```

```
      vowels += char;
```

```
    }
```

```
  } else if (/[0-9]/.test(char)) {
```

```
    numbers += char;
```

```
  }
```

```
}
```

```
console.log("Alphabets:", alphabets);
```

```
console.log("Numbers:", numbers);
```

```
console.log("Vowels:", vowels);
```

Output: Alphabets: ABCDE

Numbers: 12345

Vowels: AE

33. Globant Asked Program

```
let str = "Hello java";
let words = str.split(" ");
let result = "";
for (let word of words) {
    let reversedWord = "";
    for (let i = word.length - 1; i >= 0; i--) {
        reversedWord += word[i];
    }
    result += reversedWord + " ";
}
console.log(result.trim());
```

Output: olleH avaj